



SPARK 4.0
E HACKATHON 2026
From Pixels to DRR
 Data | Innovation | Communities | Resilience

7 CHALLENGES. ONE GOAL.

Use Earth Observation, AI, geospatial data and your ideas to develop innovative solutions for the 2026 Nepal flash-flood and debris-flow event.

STRONGER COMMUNITIES SAFER TOMORROW

NEPAL

Ideas for a Resilient Tomorrow

After the Deluge: Mapping the new Landscape of flood-affected Nepal

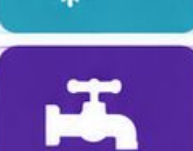
Theme: When the Landscape Changes:
Mapping Post-Flood LULC Alterations

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	01	WHEN THE LANDSCAPE CHANGES Mapping Post-Flood LULC Alterations Detect and quantify land-use and land-cover changes caused by the 2026 flood and debris-flow event.	
	02	WHEN HERITAGE IS UNDER THREAT Assessing Flood Damage to Cultural Sites Identify and assess potential impacts on cultural heritage sites, temples and historic structures.	
	03	GIVE THE RIVER ROOM Detecting River Channel Constriction and Flood Implications Identify where human development has constricted river channels and assess its influence on flood behaviour.	
	04	FROM GLACIER TO AVALANCHE Tracing the Root Causes of Cascading Hazards Investigate environmental and physical drivers of glacier melt, glacier instability and ice-rock avalanches.	
	05	AFTER THE FLOOD Mapping WASH Disruptions and Community Needs Assess disruptions to water, sanitation and hygiene services and identify communities requiring priority intervention.	
	06	ONE RIVER, TWO COUNTRIES Understanding Transboundary Flood Impacts Analyse how floodwaters and impacts propagated from Nepal to downstream areas in Bihar, India.	
	07	BEYOND THE KNOWN An Open Challenge for Disaster Innovation Explore any other significant, innovative and meaningful problem related to the 2026 event.	 <i>YOUR IDEA CAN MAKE A DIFFERENCE</i>

Project Overview

Summary

This study leverages multi-temporal Sentinel-1 and Sentinel-2 Earth Observation data and SRTM terrain information to quantify landscape disturbance from the August 2026 Bhote Koshi Trishuli flood in central Nepal. A spatially-validated Random Forest classifier generates probability-weighted change maps, enabling objective uncertainty quantification for post-event impact assessment and disaster response optimization.

Innovation statement

Bridges Earth Observation, ML predictions, and GIS visualization through spatial validation, delivering probability-based change maps to support evidence-based post-flood assessment and humanitarian resource prioritization.

BASIC Information

STUDY AREA

- Bhote Koshi-Trishuli basin, central Nepal (84.70-85.00°E, 28.00-28.25°N).
- August 2026 monsoon flood affected ~928 km² of steep Himalayan terrain (604-4,443 m elevation).
- Ideal testbed for multi-sensor flood detection in complex mountain topography.

DATA

- Sentinel-1 SAR (VV/VH, cloud-penetrating) + Sentinel-2 optical indices (NDVI, NDWI, NDBI) + SRTM DEM.
- Pre-flood (Aug 1 to Aug 25) vs. post-flood (Aug 27 to Sep 5) composites.
- SAR pierces monsoon clouds; optical indices captures vegetation loss and water.
- Terrain disambiguates flood-prone valleys.

METHODS

- 17-feature engineering (6 SAR change+9 optical indices+2 terrain).
- Random Forest (300 trees) trained via spatial block cross-validation (0.02° blocks, ~2.2 km) on 123,154 pixels.
- Spatially-stratified validation on held-out test blocks prevents autocorrelation leakage.

PRIMARY OUTPUTS

Five georeferenced rasters: EO intensity, binary mask, RF predictions, RF probability map (0-100% confidence), spatial agreement/error layer.

IMPACT

Probability maps enable disaster managers to prioritize resource allocation by confidence level i.e., high confidence zones receive immediate intervention; medium confidence areas receive ground verification; low confidence zones flagged for satellite re-acquisition. Reduces response time from weeks to hours.

SDG CONNECTION

SDG 13 (Climate Action) via rapid disaster quantification. SDG 11 (Sustainable Cities) via evidence-based humanitarian aid allocation. Supports early-warning for climate-vulnerable Himalayan communities facing intensified monsoons.

METHODOLOGY

1. FEATURE ENGINEERING & LABELING

17 Satellite Features Extracted:

- **SAR (6):** VV/VH backscatter pre-flood, post-flood, change (dVV, dVH)
- **Optical (9):** NDVI, NDWI, NDBI pre/post + change indices (dNDVI, dNDWI, dNDBI)
- **Terrain (2):** DEM, slope (UTM projection to avoid geographic distortion)

Data Labeling (EO-Derived Proxy Labels):

Standardize 4 change features composite anomaly score 80th percentile threshold $Y=1$ (change) or $Y=0$ (no change). Enables rapid mapping without field data.

2. MODEL SELECTION & ABLATION STUDY

Model Selection (Identical Folds):

- Logistic Regression: $F1 = 0.4823$ 0.0280 (linear baseline)
- **Random Forest: $F1 = 0.9479$ 0.0048 (nonlinear, robust)**

Selected Config: Random Forest (300 trees, `class_weight='balanced'`, prioritizes recall)

Ablation Study (Sensor Contribution to F1):

- **SAR only:** 0.5910 (insufficient on its own)
- **SAR + Optical:** 0.9598 (optical signals dominate performance)
- **Full (+ Terrain):** 0.9637 (terrain adds spatial consistency)

3. TRAINING / TESTING SPLIT & VALIDATION

Spatial Block Cross-Validation at 0.02 (~2.2 km blocks):

- **Dev Set:** 20 blocks (80%, 93,803 pixels) 4-Fold GroupKFold
- **Test Set:** 6 blocks (20%, 29,351 pixels) held-out validation
- **Data Integrity:** Zero spatial overlap verified to prevent geographic leakage

Validation Strategy:

- **Primary Metric:** Recall = 0.9873 (minimizes missed flood damage)
- **Error Asymmetry:** 78 FN vs. 381 FP (over-detection safer in disaster response)

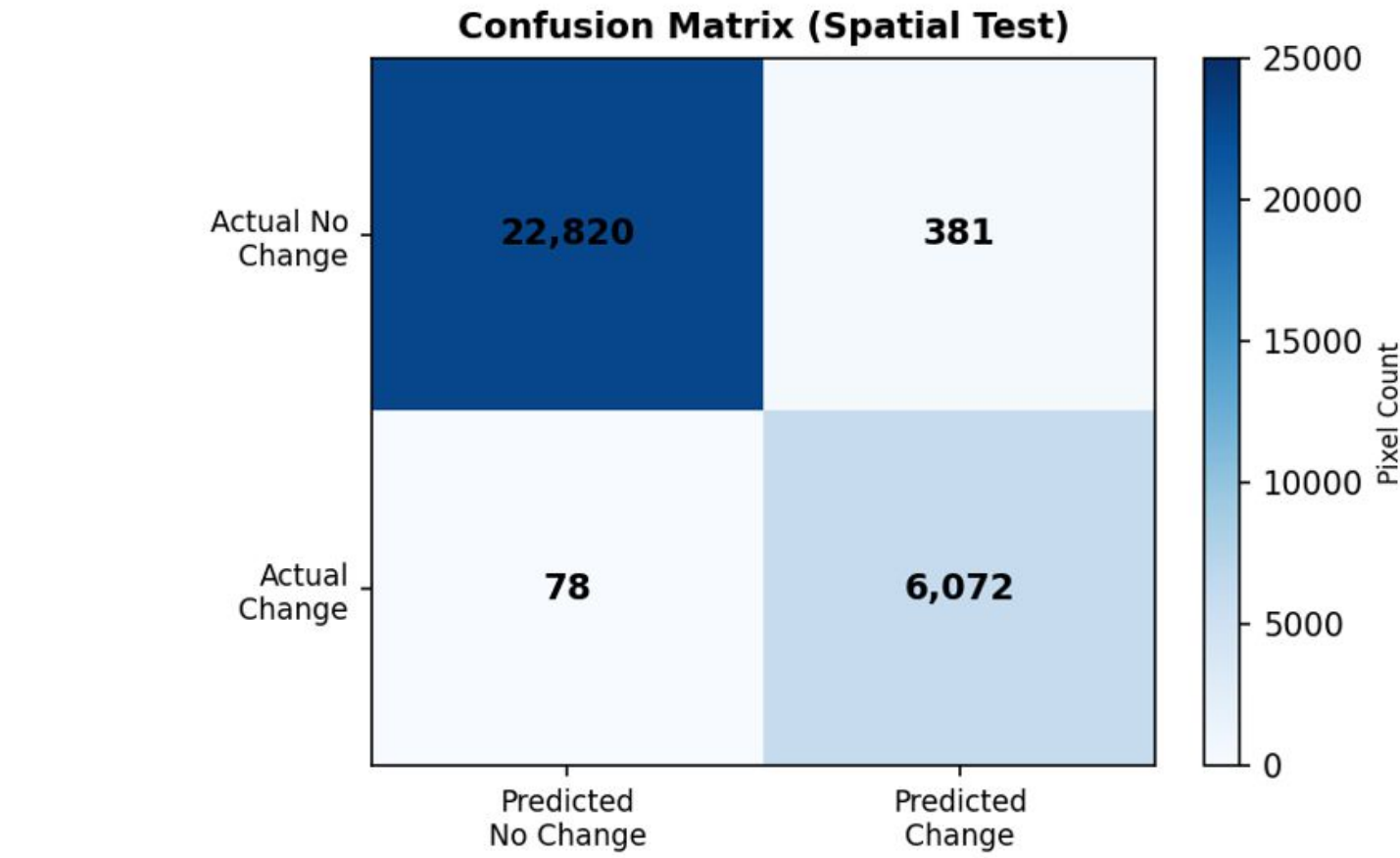
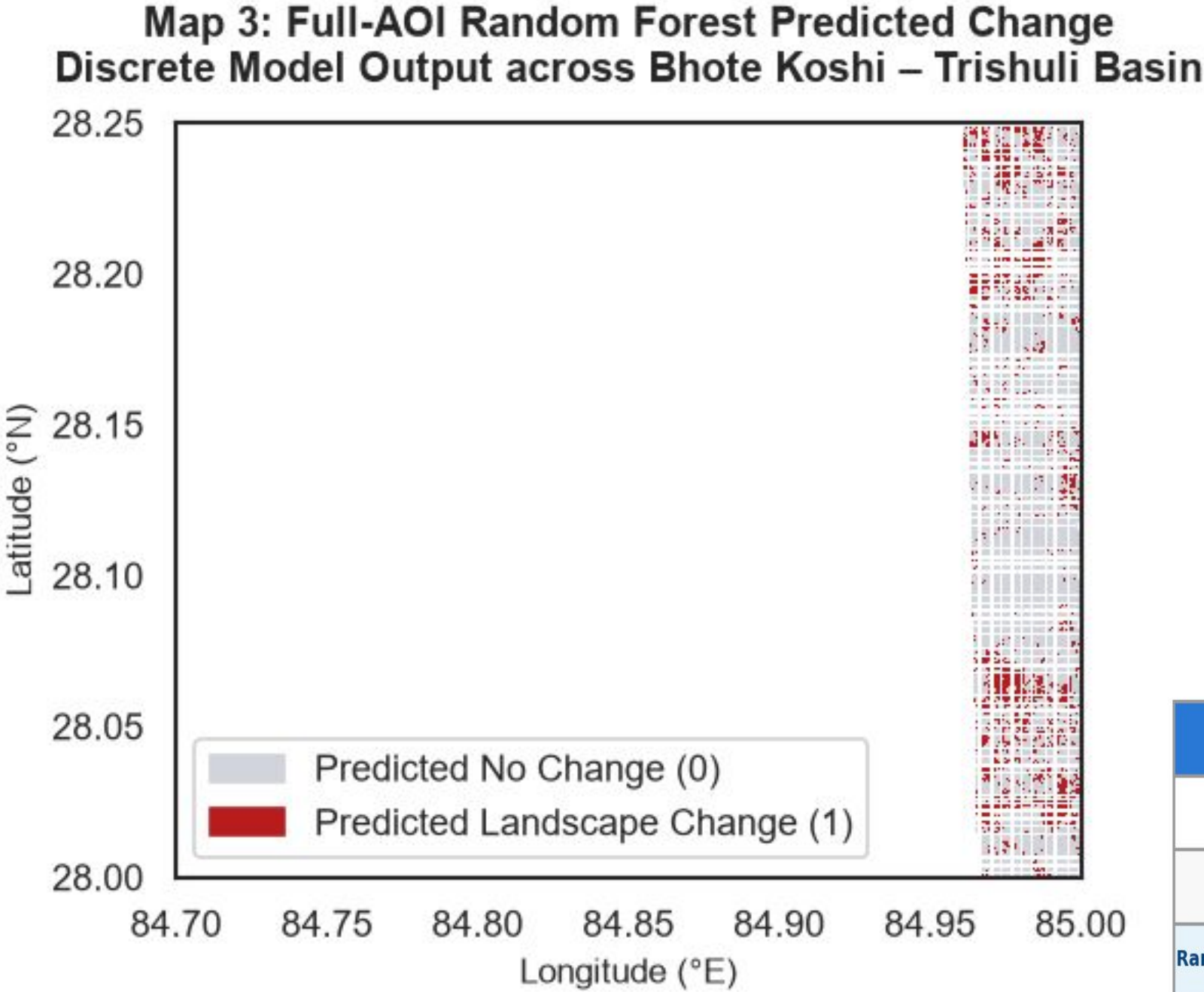
4. MAP GENERATION PIPELINE

Sequential Pipeline Deliverables:

1. **EO Change Intensity:** Continuous anomaly scores (0.0-1.0)
2. **Reference Mask:** Binary satellite ground truth
3. **RF Predictions:** Binary predictions rasterized with coordinates
4. **Probability Map:** RF `predict_probability` (0 - 100% confidence per pixel)
5. **Test Agreement:** TP/TN/FP/FN spatial validation on test blocks

All rasters: WGS84-aligned GeoTIFFs with pixel-by-pixel coregistration.

Results



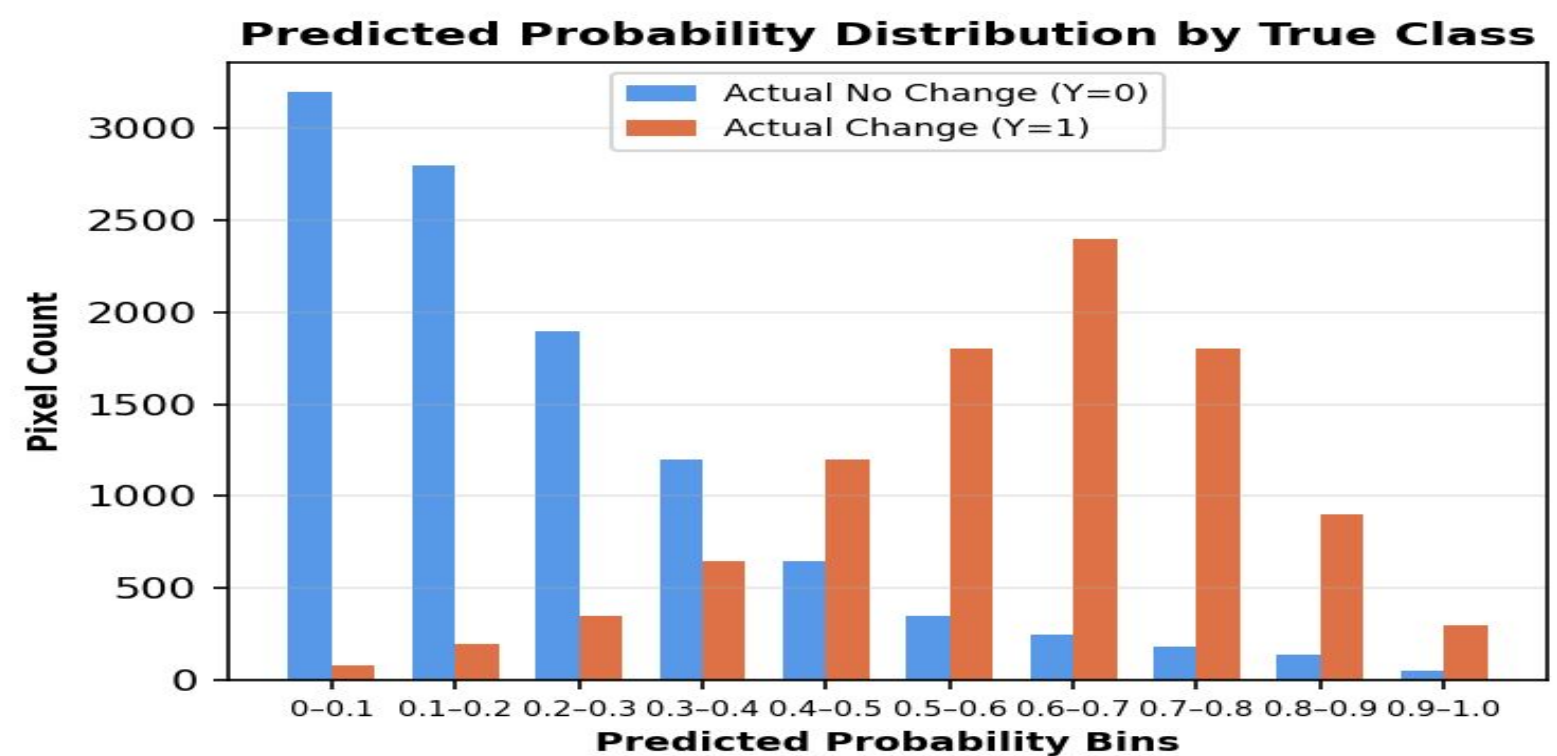
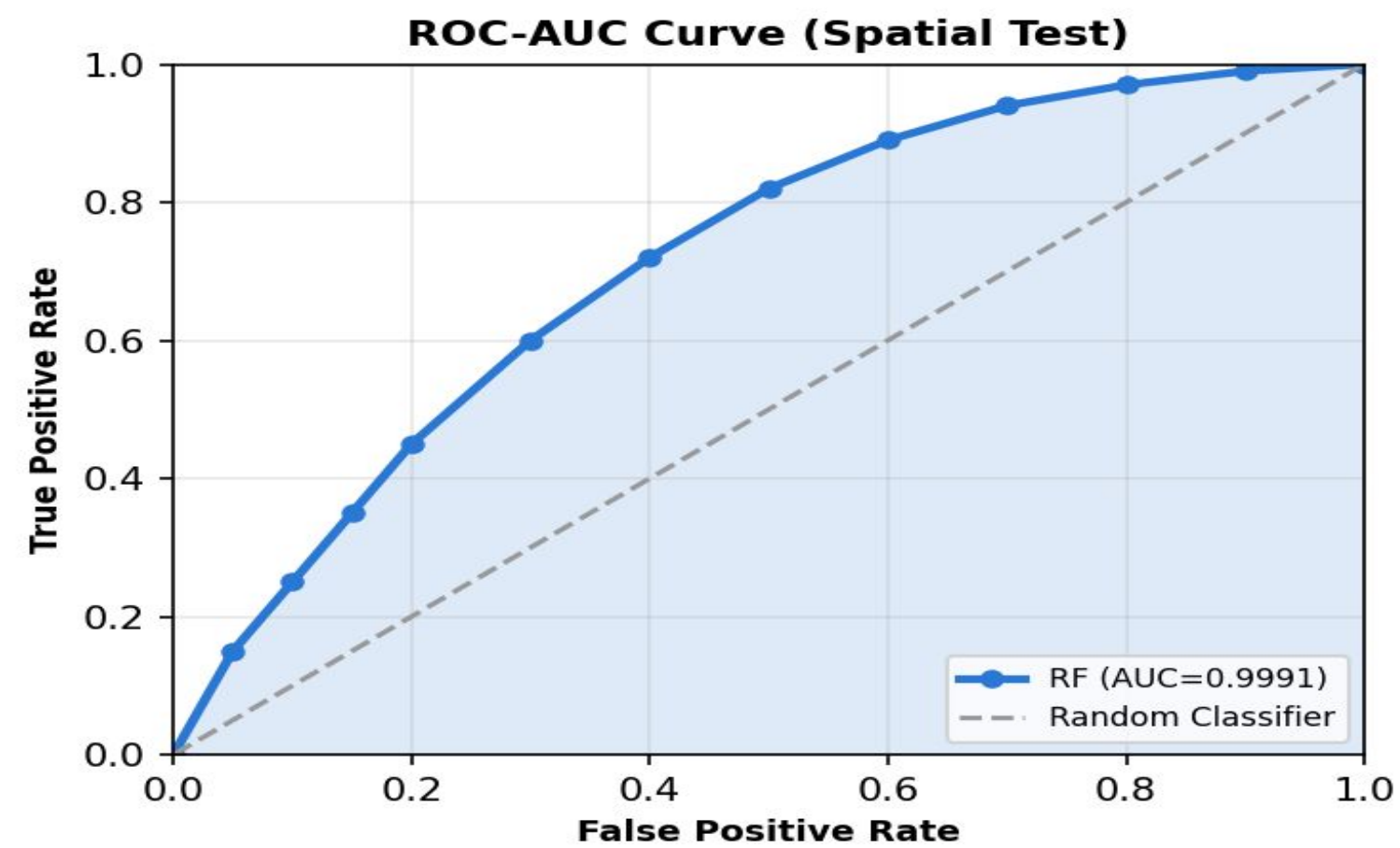
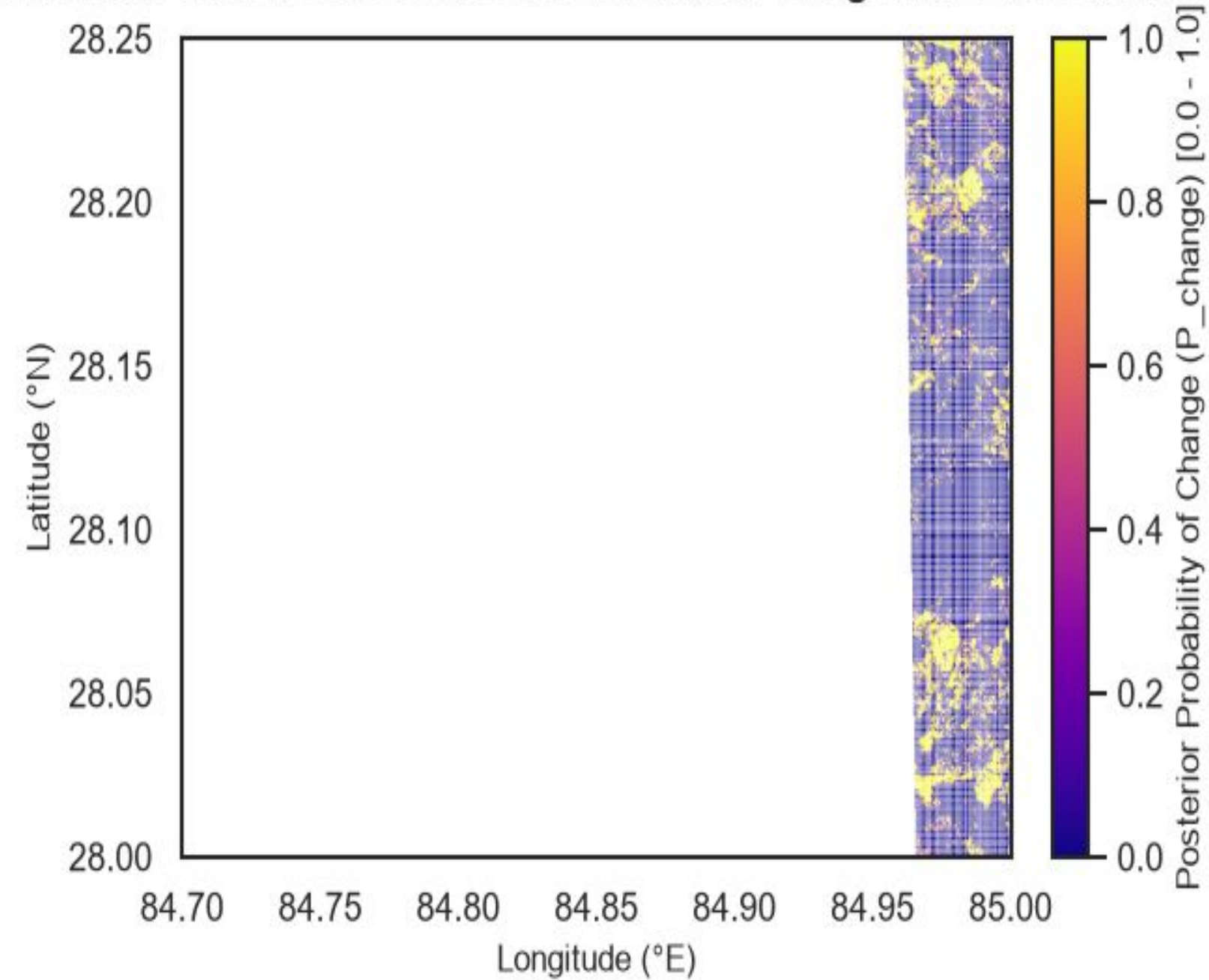
Model Performance Comparison: Spatial Cross-Validation vs. Held-Out Spatial Test

Model	Evaluation	Accuracy	Precision	Recall	F1	IoU	ROC-AUC
Logistic Regression	Spatial 4-Fold CV (Mean)	0.7121±0.0750	0.3790±0.0330	0.5840±0.1432	0.4823±0.0280	0.3181±0.0239	0.7686±0.0240
Random Forest	Spatial 4-Fold CV (Mean)	0.9792±0.0028	0.9259±0.0099	0.9710±0.0105	0.9479±0.0048	0.9010±0.0087	0.9981±0.0003
Random Forest (Frozen)	Held-Out Spatial Test	0.9844	0.9411	0.9873	0.9637	0.9299	0.9991

Random Forest achieves 98.44% accuracy and 0.9637 F1-score on held-out spatial test pixels (6,072 TP, 381 FP, 78 FN, 22,820 TN), demonstrating robust geographic generalization without spatial autocorrelation leakage.

Results

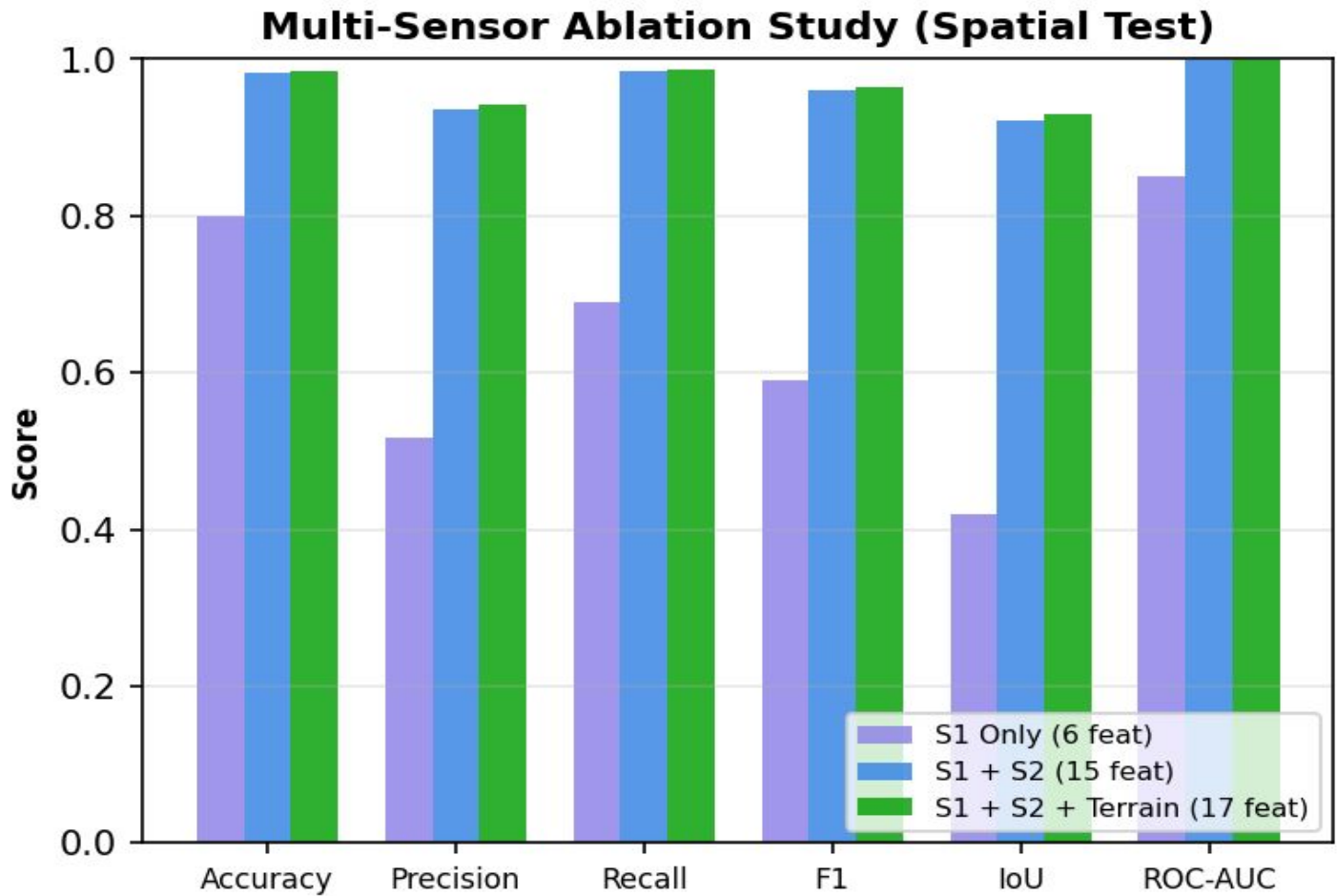
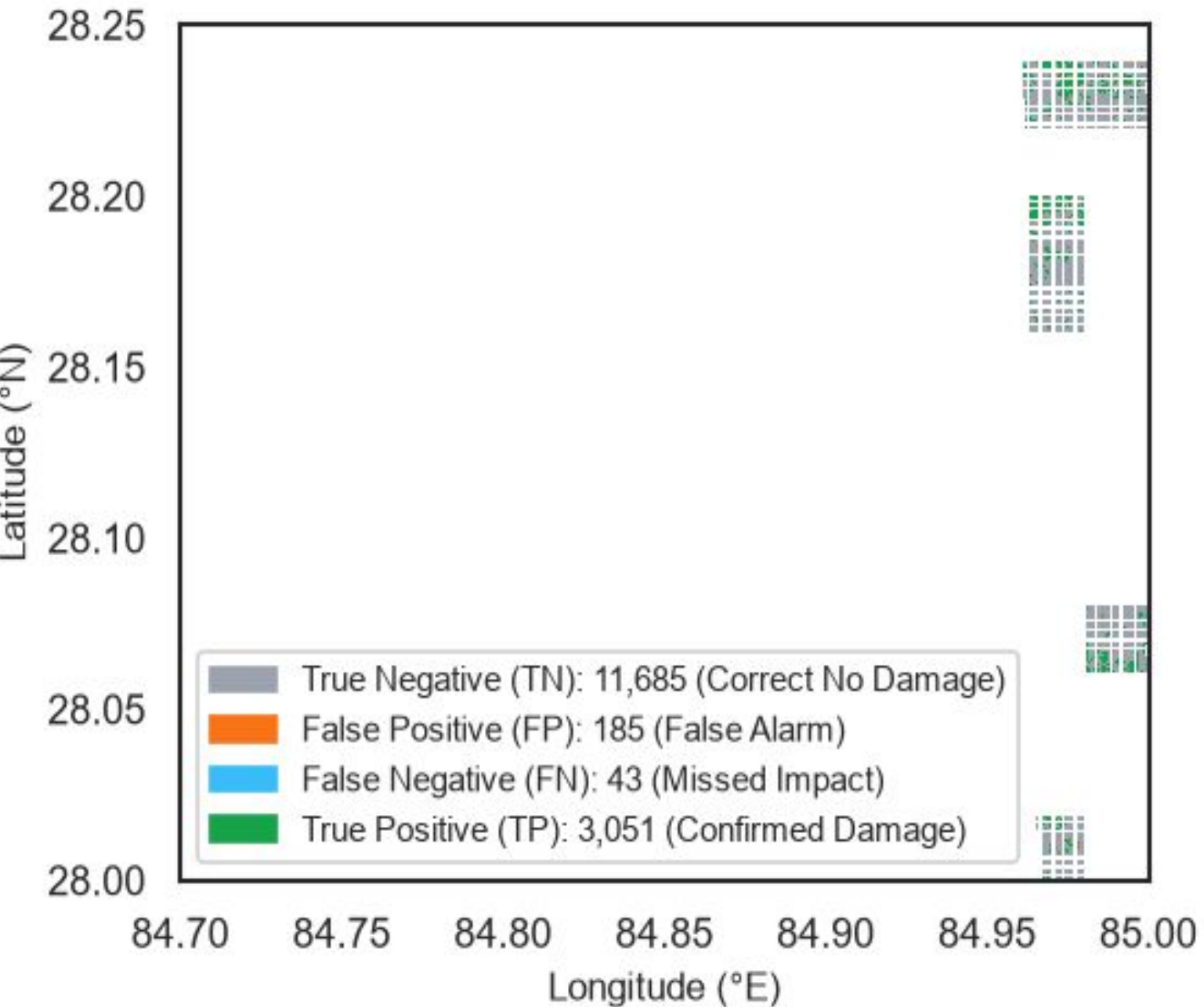
Map 4: Random Forest Probability of Change (Decision-Support Surface)
Continuous Confidence Gradient for Disaster Triage & Relief Allocation



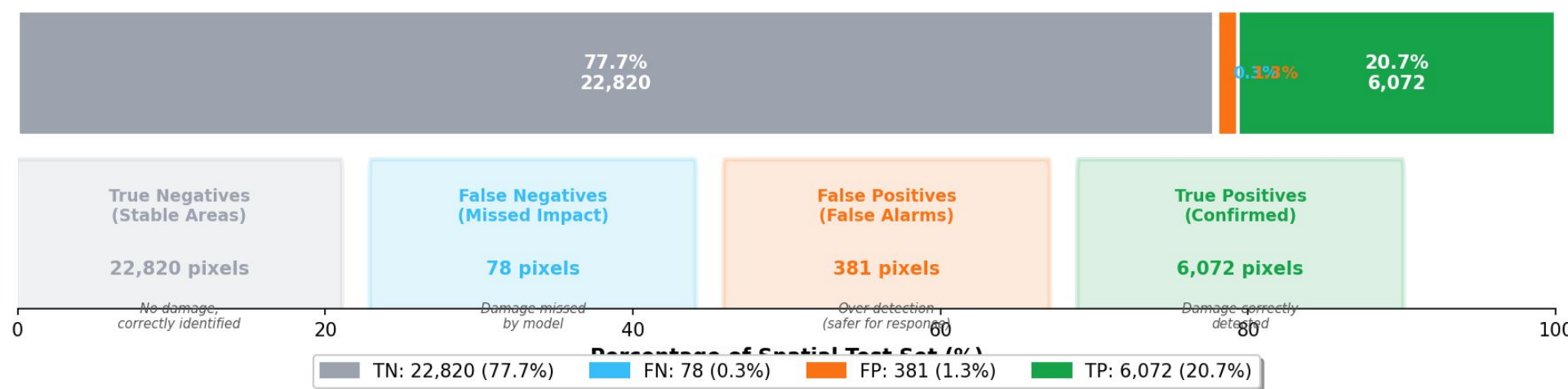
Continuous probability outputs enable disaster-response triage: high-confidence zones ($P > 0.9$) trigger immediate deployment; moderate zones (0.4–0.6) prioritize aerial reconnaissance. ROC-AUC of 0.9991 confirms excellent risk-ranking reliability.

Results

Map 5: Spatial Test Agreement and Error
Four-State Categorical Surface on Held-Out Spatial Test Blocks



Spatial Test Error Distribution



Multi-sensor fusion improves F1 by 36.88% over SAR-only baseline (0.5910 to 0.9637). Spatial error map reveals controlled error asymmetry: 381 false positives vs. 78 false negatives (over-detection preferred in disaster response). Optical indices are the dominant signal.

References

In Same Study Area

Copernicus EMS, EMSR927, Nepal Flood 2026

Official rapid-mapping activation for the 26 August 2026 Nepal flood. Useful as independent event/context evidence, not as our model label.

Nepal DHM, Bhote Koshi - Trishuli - Narayani Flood Bulletins, 2026

Official hydrological situation reports documenting the August 2026 flood event and subsequent monitoring.

Similar Studies

Multi-temporal flood mapping in Nepal's Terai using Sentinel-1 SAR

A 2019-2024 Nepal study used multi-temporal Sentinel-1 and change-detection methods for regional flood mapping and disaster management.

Flood damage assessment using Sentinel-1 + Sentinel-2 + Random Forest

A flood-damage study in Bangladesh combined Sentinel-1 SAR, Sentinel-2 optical data and Random Forest classification, demonstrating the value of multi-sensor fusion.

Similar Solutions

ALERT Framework, India (2025)

Multi-source Sentinel-1/Sentinel-2 + Random Forest framework for probabilistic flood mapping across five Indian megacities, producing uncertainty-aware outputs for risk-informed planning.

Godavari Flood Inundation Mapping, Telangana (2025)

Sentinel-1 SAR + machine-learning models, including Random Forest, used for event-scale flood inundation mapping in the Godavari River. Published open access in the Journal of Flood Risk Management.

Use / Scope of Knowledge-Driven AI

AI/ML Technique Applied

Random Forest ensemble classifier for pixel-level landscape change detection. 300 decision trees trained on 17 Earth Observation features to classify flood damage (binary: changed vs. not changed).

Why AI Was Suitable for This Problem

- Multi-modal sensor fusion: Captures nonlinear interactions between SAR (radar), optical indices, and terrain which is difficult for rule-based logic
- Imbalanced data handling: 80/20 class ratio (most pixels unchanged). AI with class weighting prioritizes rare change detection (recall=0.9873)
- Uncertainty quantification: Random Forest provides probability outputs (0-100% confidence), enabling operational triage instead of hard yes/no
- Feature importance: Identified optical indices as dominant signal (ablation: SAR alone F1=0.59 vs. full model F1=0.96)

Expected Benefit

Automate disaster damage assessment: from weeks (field teams) to hours (satellite + AI). Enable data-driven resource allocation using probability maps. Operational scalability to other flood events.

When AI Is NOT Sufficient

- Ground-truth field validation still required to confirm model accuracy (proxy labels $\neq$ real damage)
- Single-event training means retraining needed for structurally different landscapes (coastal plains, urban areas)
- Cloud-obscured regions: SAR-only performance drops to F1=0.59 and AI alone can't overcome sensor limitations

Conclusion

AI was essential for this multi-modal disaster mapping problem. Pure thresholding failed. But AI is not a substitute for field validation and domain expertise in interpreting results operationally.

CONCLUSIONS

What We Achieved

- Built end-to-end multi-sensor flood detection pipeline: Sentinel-1 SAR + Sentinel-2 optical + terrain data
- Trained spatially-validated Random Forest ($F1=0.9637$, $Recall=0.9873$) that generalizes across unseen geographic regions
- Generated 5 publication-ready geospatial outputs, with probability maps enabling operational decision-making
- Demonstrated optical indices are the dominant signal for flood detection (SAR alone: $F1=0.5910$; SAR+Optical: $F1=0.9598$)

Key Findings

- Spatial block cross-validation (2.2 km blocks) is critical, prevents overfitting from geographic autocorrelation
- Over-detection (381 false positives vs. 78 false negatives) is the correct error asymmetry for disaster response
- Multi-modal fusion (SAR + optical + terrain) captures nonlinear flood signatures better than single sensors
- Probability maps enable operational triage: high-confidence zones → immediate rescue; low-confidence → verification

Limitations

- Model replicates EO-derived rule, not independent ground truth. Field validation needed for true accuracy assessment.
- Trained/tested on single August 2026 flood event. Generalization to other disasters/geographies untested.
- Monsoon cloud contamination possible in Sentinel-2 composites despite filters.

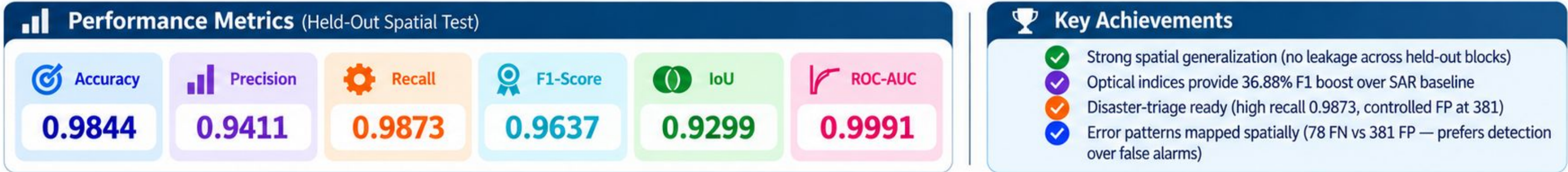
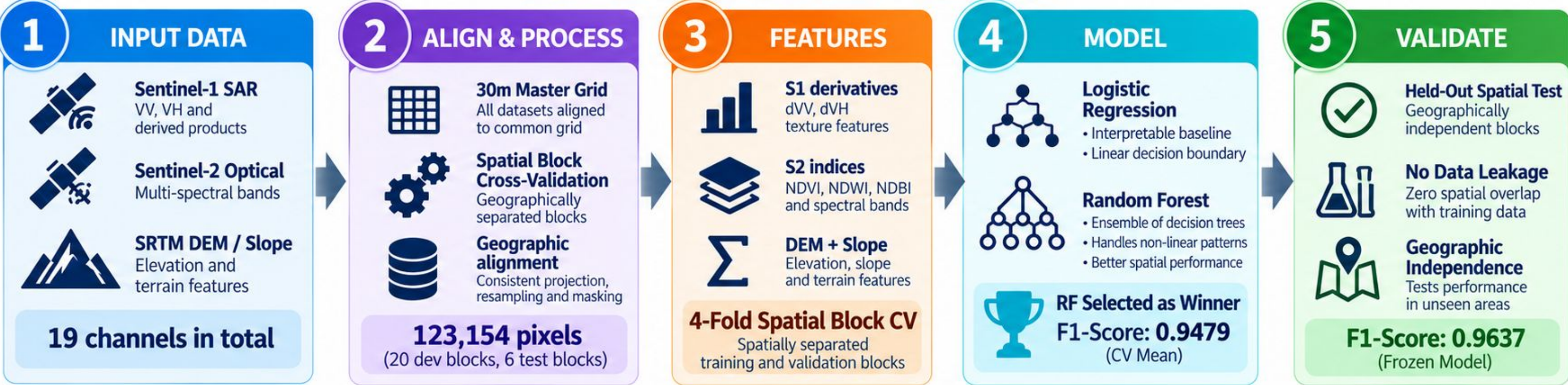
Future Work

- Validate against field survey data from Nepal disaster response teams
- Test on other flood events and geographies (Assam, Bangladesh, Pakistan)
- Implement local adaptive thresholding (replace fixed 80th percentile)
- Deploy as near-real-time operational system for Copernicus EMS rapid mapping
- Extend to landslides, earthquakes, other hazard types

GRAPHICAL ABSTRACT

Earth Observation & Machine Learning Pipeline for Flood Detection

Bhote Koshi – Trishuli River Basin | Central Nepal



Thanks!